UACE

P510/1
PHYSICS
(Theory)
Paper 1
3 Hours

ACHOLI SECONDARY SCHOOLS EXAMINATIONS COMMITTEE

Uganda Advanced Certificate of Education

JOINT MOCKS ASSESSMENT

PHYSICS

Paper 1
(Theory)
3 hours.

INSTRUCTIONS TO CANDIDATES:

*This examination paper consists of two sections; **A** and **B.** it has **six** items.*

Section **A** *has* **two** ***compulsory*** *items.*

Section B has two parts; **I** *and* **II**. *Respond to only* **one** *item in* **each** *part.*

Respond to **four** *items in all.*

Any additional item(s) responded to will **not** *be scored.*

Responses to **each** *item should begin on a* **fresh page**.

Silent non-programmable scientific calculators may be used.

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SECTION A

Item 1:

Naome, after joining secondary school, learnt about atoms and subatomic particles. They were told an atom is made up of charged particles. When she joined A'level, the subjects she takes did not clear her lingering doubts on atoms and its general properties. Meanwhile her friends in Physics class told her that they learnt about atoms and its properties and also those of its particles. For instance, a friend told her that they did an experiment where the teacher made them applied a p.d of -5000V which according to the teacher the p.d is meant to accelerate charged particles to a high velocity of $9.781\times10^{5}ms^{-1}$.

Meanwhile on another occasion, Naome's friend told her that they carried out another experiment where a spherical conductor is charged to a p.d of -50V and later connected the charged spherical conductor to a beryllium plate via an electric bulb. On irradiating the beryllium plate with light whose photon energy is 19eV, the bulb lights. On increasing the intensity of the incident radiation on the beryllium plate, the brightness of the bulb also increases. While on reducing the intensity of the incident light, the brightness of the bulb reduces. But when the beryllium plate is irradiated with light whose photon energy is less than 19eV, the bulb does not light. For radiations whose photon energy is greater than 19eV the bulb gives light, for instance when the photon energy was increased to 39eV the bulb still lit. But when the intensity of the incident radiation (whose photon energy is greater than 19eV, that is at 39eV) is made very weak, the bulb doesn't give light and yet if the beryllium plate is irradiated continuously by the weak light, after sometimes, the spherical conductor is found to be discharged.

Naome's wonder seems to be endless. One day, her brother who after completing his studies in the university returned home after a long time after having got a job. He told Naome and the family that he is working as an archaeologist; he said their work is to dig out remains of dead animals, humans and artifacts and then later determine their ages through a process called carbon dating. In one case they found an artifact made of wood and found the count rate to be 12 counts per minute.

Hints:

-motion of charged particles take place in vacuum.

-charge carried by a subatomic particle = $1.6\times10^{-19}C$.

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-Planck's constant, $h = 6.3\times10^{-34}$Js.

-count rate from a fresh piece of wood is 18 counts per minute.

Tasks:

(a) (i) Name a process for producing one of the subatomic particles.

(ii) Define the process named in (a) (i).

(b) Describe an experiment to produce a subatomic in the school laboratory.

© (i) What is the mass of the subatomic particle that is accelerated to a p.d of -5000V?

(ii)What is the charge-to-mass ratio of the subatomic particle in (d) (i)?

$\frac{1}{2}mv^2 = Ve$

(d)Explain why the;

(i) Bulb lights when the beryllium plate is irradiated with light whose photon energy is equal to 19eV.

(ii) Bulb does not light when beryllium is irradiated by light whose photon energy is less than 19eV.

(iii) The brightness of the bulb changes when the intensity of the radiation incident on the beryllium plate changes.

(iv) Spherical conductor get discharged when the beryllium plate is irradiated by light whose photon energy is greater than 19eV during which the bulb doesn't light.

(e) (i) Find the frequency of the light just enough to cause the bulb to light.

(ii) During the process when the beryllium plate is irradiated by light whose photon energy is greater than 19eV, that is 39eV, what is the speed of the emitted particle from the beryllium surface?

(f) (i) (h) (i) Determine the age of the artifact.

(ii) What are the limitations of carbon dating in the modern world?

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Item2:

When Akello was in S.2. One day during physics lesson, they carried out an experiment to determine what factors affects the period, T of oscillation of a simple pendulum. They found that the period, T of a simple pendulum depends on the length, l of the thread of the pendulum. They also found that on changing the mass of the pendulum bob to different masses, they found that the period of oscillation of the bob does not change provided the length of the thread is kept constant. And their teacher also told them that the period of oscillation of a simple pendulum depends on the value of acceleration due to gravity at a particular place. The teacher said for instance, the value of the acceleration due to gravity on the moon's surface is $\frac{1}{6th}$ the value on earth's surface. That means the period of oscillation, T_m of a simple pendulum on moon's surface is 6 times the period of oscillation T_E of a simple pendulum of the same length on earth's surface. I.e. $T_m = 6T_E$. and the teacher was saying the field the pendulum oscillate in on moon's surface is different from that on earth's surface.

While in school, Akello is the school goalkeeper. Always while catching a ball, she has to extend her arms forward and start catching the ball from an arm's length and then draws the ball towards her chest. Whenever she catches the ball without drawing it towards her chest, the ball always rebounded back from her grip. She doesn't understand why. One time she caught a ball of mass 250g moving horizontally at $12ms^{-1}$ and pull it backwards until it came to rest on her chest.

Hint:

On earth's surface the value of acceleration due to gravity, $g = 9.81ms^{-1}$.

Task:

As someone who has studied physics up to A 'level help Akello improves her understanding in some of the areas she has doubts in order to clear her doubts.

(a) (i) show that the period of oscillation of the pendulum is given as;

$$T = 2\pi\sqrt{\frac{l}{g}}$$

(ii) Name the method used to obtain the relation in (a) (i).

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(b) (i) Explain why the period, T of oscillation of a simple pendulum does not depend on the mass of the pendulum.

(ii) Prove the formula; $T_m = 6T_E$.

(d) As the pendulum oscillates from side to side, what type of field does it oscillate in when on;

(i) Earth's surface.

(ii) Moon's surface.

(iii) Mention any characteristic differences that exist between the types of field mentioned in (d) (i) and (ii).

(e) State the principle that applies to the oscillation of a simple pendulum particularly when on moon's surface.

(f) Explain the following to Akello;

(i) Why she has to pull the ball back towards her chest.

(ii) Why when she doesn't pull her arm back the ball rebounds back.

(g) Find;

(i) How long it took the ball to come to rest.

(ii) The average force exerted by Akello in catching the ball.

$V = u + at$

$S = ut + \frac{1}{2}gt^2$

$V^2 = u^2 + 2as$

$\sqrt{y} = v_t =$

SECTION B

PART I

Item 3

Bright is an S.1 learner. One day during physics lesson their teacher told them that light travels in a straight line. And yet to bright, he sees a flood of light filling space all around him. For instance to him he sees a flood of light coming from a torch, a firefly etc. The teacher also told him that we see things around us by the light they reflect to

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our eyes. One day while playing with a plane mirror he realizes that the plane mirror throws a lot of light from the sun onto the wall. But when he tries the back side of the plane mirror to reflect sunlight onto the wall, no light is reflected onto the wall. He wonders why the backside of the mirror doesn't reflect light and yet the front page does. But since we are able to see both the front face of the mirror and its backside it means they both reflect light. At the moment when the front face of the mirror is reflecting light onto the wall, he realizes that any slight shaking of his hand is shown by the movements of the reflected beam of light on the wall. It seems as if the shaking of the beam of light on the wall amplifies his hand's shaking.

One day Bright and his classmates together with their teachers visited a park. One of their teachers was carrying a type of telescope called a prism binocular. In the park they used it to view animals such as elephants, lions, giraffe etc. their physics teacher was trying to tell them that it has a pair of objective lenses each of focal length, f_o and also a pair of eyepiece lenses each of focal length, f_e. And the teacher also told them that the effective length of the telescope is 15cm. while the focal length of each eyepiece lens is 5cm. the teacher was saying there are other types of telescopes such as astronomical telescope, reflecting telescope etc.

All these explanation about the telescope Bright has no understanding of at all.

Hint: the effective length of the prism binocular = $\frac{1}{3}$ the effective length of a refracting astronomical telescope of same power.

Tasks:

As someone who has learnt physics at A' level try to make Bright understand in areas where he has doubts.

(a) Name the flood of light coming from the following;
(i) A torch and illustrate it diagrammatically.

(ii) A fire fly and illustrate it diagrammatically.

(b) Illustrate and explain how reflection from the following takes place;
(i) The front surface of a plane mirror.
(ii) The backside of the mirror.

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(c) As Bright hand shakes, the reflected beam shakes much more. Show how the shaking of Junior hand is related to the shaking of the reflected beam on the wall.

(d) Name where the relationship you have shown in part © is applied, stating its advantages.

(d) (i) Explain with diagrammatical illustration, how a prism binocular enable people see objects.

(ii) Obtain an expression for the relationship between f_o and f_e of a prism binocular.

(iii) Find the best position of the eye for the prism binocular.

(d) State one;

(i) Advantage of the prism binocular over other types of telescope.

(ii) Disadvantage of the prism binocular over other types of telescope.

Item 4.

Learners in S.5 put 0.5kg of water in a boiler and started heating it using electric heater through which a current of 23.5A is passed and a p.d of 200V is applied. Before switching on the current they placed a thermometer in the liquid and noted that the initial temperature of the water is19°C. After switching on current, the reading of the thermometer kept on rising. After five minutes the reading of the thermometer stopped rising and remained steady at θ°C. Three minutes after the reading of the thermometer has stopped rising, the mass of the water in the boiler reduced by 324g. One of the learner suggested that they should fasten the process of reducing the mass of the water in the boiler. As a result, they increased the current through the heater to 30.0A and the p.d to 250V. After two minutes they lost 349.1g of water.

Hint: the specific heat capacity of water = $4200Jkg^{-1}K^{-1}$.

Task:

(a)(i) Explain why the temperature of the water remained constant at θ°C.

(ii) What is the heat supplied at constant temperature called?

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(b) Calculate the value of the

(i) Specific latent heat of vaporization of water.

(ii) Heat lost to the surrounding.

(iii) mention two sources of errors in obtaining the value obtained in (b) (i).

© (ii) Draw a diagram that can be used to obtain a more accurate result for the quantity named in (a) (i).

(iii) Describe how the diagram drawn in (b) (ii) can be used to obtain a more accurate result.

SECTION B

PART II

Item 5.

Junior a learner in O'level found some S.5 learners doing physics practical. At first they connected 20cm of a wire mounted on a meter rule in series with an ameter, a dry cell in a cell holder and a switch. On closing the switch the galvanometer gets deflected and remained at its maximum position. When Junior asked them as to why the ammeter gets deflected when the switch is closed, they told him because it shows that electric current is flowing in the circuit. They said that electric current is due to the movements of electric charge in the wire carrying current. They said the electric current moves at an average speed in the conductor and their movements is caused by the dry cell. But when they changed the length of the wire, say to 30cm, the current also changes as indicated by the ammeter. At an instance when the length of the wire, l = 20cm, they connected a voltmeter across the terminals of the cell. When the switch is closed the voltmeter reads 1.25V and when they opened the switch, the voltmeter reads 1.5V. When Junior asked them as to why the reading of the voltmeter drops when the switch is closes. They told him it is because of internal resistance in the cell. They then closed the switch and changed the length of the wire to l = 30, 40, 50, 60 and 70cm and each time noted the reading of both the voltmeter and the ammeter and then entered their values into a table of values and later drew a graph of current against voltage.

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Task:

As someone who has studied physics at A'level try to make junior understand what the S.5 learners did by doing the following.

(a)(i))what causes variation in the amount of current when the length of the conductor is varied?

(ii) Obtain an expression for the uniform speed with which the moving charges that constitute current in the wire move.

(b)(i) Draw the circuit diagram the learners connected.

(ii) Draw the sketch of the graph the learners drew.

(iii) What is the significance of the graph?

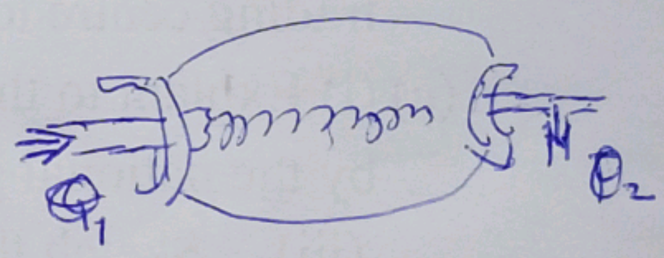
(iv) State the law that is depicted by the graph.

(v) Conductors whose graphs are drawn in (b) (ii), what are they called?

(vi) Why does the reading of the voltmeter drop when the switch is closed?

© (i) if the resistivity of the wire is $4.9\times10^{-7}\Omega m$ and its diameter is 0.64mm, what is the value of the current when the length of the wire, l = 20cm?

(ii) What is the internal resistance of the cell?



Item 6.

A learner in O'level lives in a village that is not connected to the national grid. One time the government decided to connect the closest town to their village to the national grid and the power line passes via their village. He saw the electric cables use for carrying electricity are not coated. While in the nearby trading centre, the inhabitant of the centre wanted power. They were told to pay for a transformer which should be installed before they can use the electricity. After installing the transformer, power was connected in people's houses using electric wires that are coated by a plastic kind of material which they say is an insulator. While the cables themselves are conductors. This learner with limited knowledge in electricity doesn't know the difference between conductors and insulators. The learner also wonders if this kind of electricity supplied

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by the national grid is the same as those supplied by accumulators that are charged by solar panels which some people in their village use for charging their phones. The people who installed the transformer talked of the electricity in the national grid as having what they called the root-mean-square value. The people who installed the transformer were saying that the voltage of the national grid is 11kV and the current is 1.136A. While the voltage from the transformer is 250V and they were saying the transformer is 96% efficient.

Task:

As someone who studied physics ion A 'levels, you let the boy understand some basic ideas on electricity. The boy on electricity by answering the following questions:

(a) (i) Explain why the electric wires used for distributing electricity in people's houses were coated.

(ii) Explain why some materials are able to conduct electricity while others don't.

(iii) Describe an experiment to distinguish between the cables used and their coatings in the scenario above.

(b) Describe how the transformer installed was used to connect the inhabitants of the trading centre to the national grid.

(c) (i) Explain to the learner the difference between the kind of electricity supplied by the national grid and those the villagers use for charging their phones.

(iii) Sketch the graph of the type of electricity available on the national grid.

(d) Show how a root-mean-square value is obtained.

(e) Calculate the value of the current that the transformer supply in the trading centre.

THE END.

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